

Application of Saturation Constraints to the Lion MRAC Scheme

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1. Plant

The dynamic system considered in this experiment is a second-order plant parameterized as

$$a_0 \ddot{y}(t) + a_1 \dot{y}(t) + b_0 y(t) = u(t)$$

where the coefficients for Variant 7 (my variant) are:

$$a_0 = 1, a_1 = 3, b_0 = 3.$$

This places the open-loop poles in the left half-plane, resulting in a stable but relatively slow and oscillatory baseline plant. The plant is assumed to be fully actuated, and all its states (and their needed estimates) are available for measurement or computation. The system is assumed to satisfy the necessary conditions for model reference adaptive control (MRAC), including controllability and observability. No explicit disturbances or unmodeled dynamics are included, focusing the analysis on the parametric uncertainties in a_0 , a_1 , and b_0 .

2. Control Objective

The goal is to synthesize an adaptive controller such that the plant output $y(t)$ tracks the trajectory of a prescribed reference model $y_m(t)$, regardless of the uncertainty in the plant parameters. Mathematically, the objective is

$$\lim_{t \rightarrow \infty} |y(t) - y_m(t)| = 0$$

under suitable excitation of the reference. Additionally, the MRAC scheme should drive the parameter estimates to their true values as

$$\lim_{t \rightarrow \infty} \|\hat{\theta}(t) - \theta^*\| = 0,$$

where $\hat{\theta}$ collects the adaptive parameter vector. The design must ensure signal boundedness and robust stability of the closed-loop system throughout operation.

3. Reference Model

The chosen reference model is second-order and strictly stable,

$$K_M(s) = \frac{1}{s^2 + 2s + 1},$$

so in the time domain it is described by

$$\ddot{y}_m + 2\dot{y}_m + y_m = g(t),$$

where $g(t)$ is the external reference input. With double poles at $s = -1$, this reference acts as a low-pass system, producing smooth output even when the input undergoes moderate discontinuity. Two reference signals were used: the primary,

$$g(t) = \text{sign}(\sin t) + 2$$

(which switches sharply between 1 and 3), and a secondary,

$$g(t) = \sin t + 2$$

(which is smooth), to test the importance of reference signal continuity and richness for adaptive convergence.

4. Solution: Lion MRAC Algorithm with Saturation Constraints

The Lion scheme augments the traditional gradient MRAC law by using dynamic regressor extension and normalization to achieve exponential parameter convergence.

Properties and Assumptions of the Lion Scheme

The Lion scheme exhibits the following fundamental properties and assumptions necessary for its practical and theoretical guarantees:

1. **Bounded Parametric Errors:** The vector of parametric estimation errors remains bounded. Specifically, if the adaptive parameters $\psi(t)$ and the constructed signals $Y(t)$ are bounded, then the deviation $\tilde{\theta}(t) = \psi(t) - \theta^*$ from the true parameters is bounded.
2. **Asymptotic Stability:** Under boundedness of $\psi(t)$ and $Y(t)$, the tracking error $e(t) = y_m(t) - y(t)$ and the parameter estimation error $\tilde{\theta}(t)$ converge asymptotically to zero as time tends to infinity.
3. **Exponential Convergence Under Persistent Excitation (PE):** The parameter error $\tilde{\theta}$ converges *exponentially* if and only if the extended regressor vector $W(t)$ satisfies the persistent excitation condition. Formally, there exists $\alpha > 0$ such that for all $t \geq 0$,

$$\int_t^{t+T} W(\tau) W^\top(\tau) d\tau \geq \alpha I,$$

where $T > 0$ is a finite excitation interval and I is the identity matrix.

4. **Relaxed PE Condition:** Even if the smallest eigenvalue $\lambda_{\min}$ of the excitation matrix $\int_t^{t+T} W(\tau) W^\top(\tau) d\tau$ approaches zero asymptotically, asymptotic convergence of the error is preserved.
5. **Gain Normalization:** The adaptation gain γ is dynamically adjusted according to the magnitude of the regressor norm, ensuring monotonic and element-wise exponential convergence,

$$\gamma(t) = \gamma_0 \frac{1}{1 + \|W(t)\|^2},$$

where $\gamma_0 > 0$ is a constant scalar gain.

6. **Requirements on Filters $H_i(s)$:** The filters used in dynamic regressor extension,

$$H_i(s) = \frac{d_i}{s + d_i}, \quad d_i > 0,$$

must be proper, stable, minimum phase, and asymptotically stable. The distinct filters $H_i(s)$ produce filtered regressors $\phi_{\text{bar},fi}$ that are linearly independent, enabling improved excitation richness.

The state regressor is

$$\omega = \begin{bmatrix} y \\ \nu_1 \\ \nu_2 \end{bmatrix}$$

where

$$\dot{\nu}_1 = -\lambda\nu_1 + y, \quad \dot{\nu}_2 = -\lambda\nu_2 + u.$$

5. Assumptions

The theoretical validity and practical performance of the Lion MRAC algorithm hinge upon a set of fundamental assumptions concerning the plant model, the reference signal, the filters used for dynamic regressor extension, and the nature of the unknown parameters. These assumptions are essential for guaranteeing the stability and convergence properties outlined previously.

1. **Plant Model Structure:** The controlled plant is assumed to be accurately represented by a linear parametric model. For the specific second-order plant under consideration (Variant 7), this implies that after applying appropriate filters (e.g., related to the polynomial $K(s)$ used in the MRAC design), the relationship between the plant output y , input u , and the unknown parameters can be expressed linearly as $y = \psi^{*T} \phi_p + \delta(t)$, where ψ^* is the true vector of constant parameters, ϕ_p is the regressor vector containing filtered versions of y and u , and $\delta(t)$ represents terms ideally negligible or handled separately (e.g., via the augmented error mechanism).
2. **Unknown Parameters:** The vector of unknown parameters ψ^* (encompassing coefficients like a_0, a_1, b_0 in the plant model) is assumed to be constant over the time scale of the adaptation process.

3. **Filters for Dynamic Regressor Extension (DRE):** The transfer functions $H_i(s)$ employed in the DRE step must satisfy specific criteria to ensure the effectiveness of the extended regressor $W(t)$. These filters, often chosen as $H_i(s) = \frac{d_i}{s+d_i}$ or $H_i(s) = \frac{1}{s+d_i}$ (where $d_i > 0$), must be:
 - **Proper:** The degree of the denominator polynomial must be greater than or equal to the numerator polynomial.
 - **Asymptotically Stable:** All poles of $H_i(s)$ must lie in the left half of the complex plane (i.e., $d_i > 0$). This ensures that the filtered signals remain bounded for bounded inputs.
 - **Distinct Poles:** The poles d_i of the filters must be distinct ($d_i \neq d_j$ for $i \neq j$). This distinctness is crucial for ensuring that the resulting filtered regressors $\omega_i(t)$ are linearly independent, thereby enriching the information content of the extended regressor matrix $W(t)$.
 - **Minimum Phase (often):** While not always strictly necessary for basic stability, it is common for the filters $H_i(s)$ to be minimum phase (all zeros in the left half-plane or on the imaginary axis) to facilitate analysis and robustness.
4. **Signal Boundedness:** The reference input signal $g(t)$, the resulting plant output $y(t)$, the control input $u(t)$, the regressor vector $\omega(t)$, and the filtered signals (like Y_i , $\phi_{\text{bar},fi}$) are assumed to remain bounded during the operation of the closed-loop system. This assumption is necessary for the subsequent stability analysis.
5. **Reference Signal Richness (for Convergence):** While the algorithm ensures boundedness under the above assumptions, convergence of the parameter estimates $\hat{\psi}(t)$ to the true values ψ^* requires the extended regressor matrix $W(t)$ to be persistently exciting (PE). The richness of $W(t)$ depends on the excitation properties of the original regressor $\omega(t)$ and the distinct filters $H_i(s)$. The reference signal $g(t)$ (and the resulting $y(t), u(t)$) must provide sufficient information content to excite the system dynamics adequately.

6. Derivation

The derivation of the Lion MRAC adaptation law proceeds from a fundamental static parametric error model by employing the technique of Dynamic Regressor Extension (DRE). This process transforms the single error equation into a system of equations, enhancing the excitation properties and enabling the specific structure of the Lion adaptation law.

Step 1: Static Parametric Error Model

The derivation begins with a foundational relationship linking an error signal $\varepsilon(t)$ to a regressor vector $\omega(t)$ through the vector of unknown constant parameters θ^* . In the context of MRAC, this often arises from an augmented error formulation. Let $\varepsilon_a(t)$ represent an augmented error signal (related to $y - y_m + \hat{\psi}^T \phi$). The static model is:

$$\varepsilon_a(t) = \omega^T(t)\theta^*$$

where $\omega(t)$ is the regressor vector (e.g., $\omega = [y, \nu_1, \nu_2]^T$ as defined in the report) and θ^* represents the true unknown parameters (or a part thereof, like ψ^* in the report's notation).

Step 2: Dynamic Regressor Extension (DRE)

To enhance the excitation properties, q distinct, proper, asymptotically stable transfer functions $H_i(s)$ are applied to both sides of the static error model. For linear operators applied to signals multiplied by constant parameters, this yields:

$$H_i(s)[\varepsilon_a(t)] = H_i(s)[\omega^T(t)\theta^*]$$

$$Y_i(t) = (H_i(s)[\omega(t)])^T \theta^*$$

Define the filtered regressor vector as $\omega_i(t) = \mathcal{L}^{-1}\{H_i(s)[\omega(t)]\}$. Therefore,

$$Y_i(t) = \omega_i^T(t)\theta^*$$

Step 3: Stacking the Extended Model

The q equations obtained in Step 2 are stacked to form an extended model:

$$Y(t) = W(t)\theta^*$$

where:

- $Y(t) = \begin{bmatrix} Y_1(t) \\ Y_2(t) \\ \vdots \\ Y_q(t) \end{bmatrix}$ is the vector of filtered outputs/errors.
- $W(t) = \begin{bmatrix} \omega_1^T(t) \\ \omega_2^T(t) \\ \vdots \\ \omega_q^T(t) \end{bmatrix}$ is the extended regressor matrix, with rows being the transposed filtered regressors.

Step 4: Formulating the Adaptation Law

The objective is to estimate the unknown parameter vector θ^* . The Lion scheme proposes an adaptation law based on the extended model $Y = W\theta^*$. A common approach (motivated by applying gradient or similar principles to the extended model) is to define the rate of change of the parameter estimate $\hat{\theta}(t)$ proportional to the product of the extended regressor and an extended error. The extended error vector $E(t)$ is defined as the difference between the actual filtered signals $Y(t)$ and the signals predicted by the current estimate $W(t)\hat{\theta}(t)$:

$$E(t) = Y(t) - W(t)\hat{\theta}(t)$$

The Lion adaptation law is then formulated as:

$$\dot{\hat{\theta}}(t) = \gamma W^T(t)E(t)$$

where $\gamma > 0$ is the adaptation gain. Substituting the definition of $E(t)$:

$$\dot{\hat{\theta}}(t) = \gamma W^T(t)(Y(t) - W(t)\hat{\theta}(t))$$

Expanding this yields the standard form:

$$\dot{\hat{\theta}}(t) = \gamma W^T(t)Y(t) - \gamma W^T(t)W(t)\hat{\theta}(t)$$

In the specific implementation detailed in the report (and in the associated MATLAB code), this translates to the sum of terms involving the individual filtered regressors $\phi_{\text{bar},fi}$ (representing ω_i) and their corresponding filtered errors E_i :

$$\dot{\psi} = \gamma_{\text{norm}} (\phi_{\text{bar},f1} E_1 + \phi_{\text{bar},f2} E_2 + \phi_{\text{bar},f3} E_3)$$

where $\gamma_{\text{norm}} = \gamma_0 / (1 + \|\omega_{\text{bar}}\|^2)$ is the normalized adaptation gain.

Step 5: Parameter Error Dynamics (for Analysis)

To analyze the convergence of the parameter estimates, the dynamics of the parameter error $\tilde{\theta}(t) = \theta^* - \hat{\theta}(t)$ are derived. Taking the time derivative:

$$\dot{\tilde{\theta}}(t) = \dot{\theta}^* - \dot{\hat{\theta}}(t) = 0 - \dot{\hat{\theta}}(t) = -\gamma W^T(t)(Y(t) - W(t)\hat{\theta}(t))$$

Substituting the true relationship $Y(t) = W(t)\theta^*$:

$$\dot{\tilde{\theta}}(t) = -\gamma W^T(t)(W(t)\theta^* - W(t)\hat{\theta}(t))$$

$$\dot{\tilde{\theta}}(t) = -\gamma W^T(t)W(t)(\theta^* - \hat{\theta}(t))$$

$$\dot{\tilde{\theta}}(t) = -\gamma W^T(t)W(t)\tilde{\theta}(t)$$

This differential equation governs the evolution of the parameter error and is central to the Lyapunov-based stability and convergence analysis. If the extended regressor $W(t)$ is persistently exciting, the matrix $W^T(t)W(t)$ provides sufficient “negative feedback,” ensuring that $\tilde{\theta}(t)$ converges to zero, often exponentially.

The corresponding control law (for plant invertibility and reference matching) is:

$$u = \frac{1}{\hat{b}_m} (-\hat{\theta}^\top \omega + a_{m0} g)$$

where $\hat{b}_m$ is the adaptive estimate of the plant’s input parameter, and a_{m0} is the trailing coefficient of the model.

Also, the normalization in the adaptation law adjusts the adaptation gain when the regressor norm is large, preventing instability and improving convergence under persistent excitation. The dynamic regressor extension (with three different filters) creates a richer excitation signal, a prerequisite for guaranteeing not just stability but exponential convergence of both the tracking and parameter error.

7. Results

7.1 Tracking Error $\varepsilon(t) = y_m(t) - y(t)$

The tracking error for both the unsaturated and saturated cases is shown in the following plots.

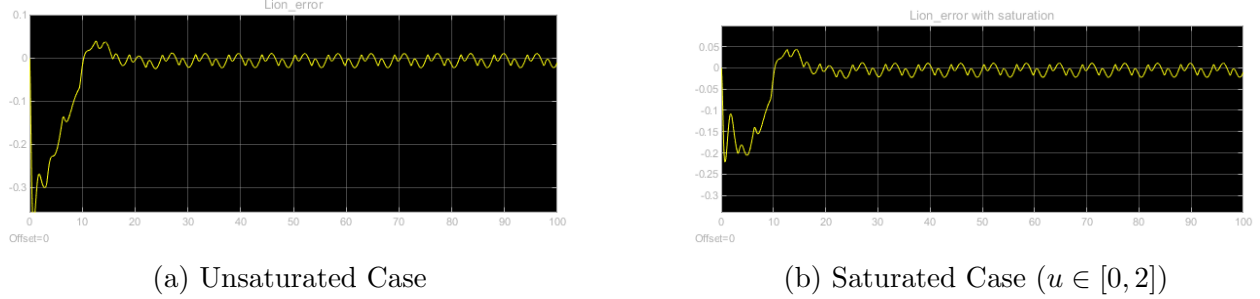


Figure 1: Tracking error for Lion MRAC with and without saturation.

Observation:

In the **unsaturated case**, the error converges rapidly within the first 10–15 seconds to a small oscillatory band around zero, with peak-to-peak amplitude approximately ± 0.08 .

In the **saturated case**, the error also converges quickly in the initial phase (first 10–15 seconds), but it settles into a steady-state oscillation around zero with a slightly smaller amplitude (± 0.05).

This indicates that *the Lion scheme achieves rapid convergence in both cases*, and the saturation constraint does not prevent convergence unlike the Gradient scheme.

The reference signal $g(t) = \text{sign}(\sin t) + 2$ requires the plant to switch between 1 and 3. To track this, the controller must generate a high-amplitude control signal which frequently exceeds the saturation limit of 2.0.

When $U_s(t) > 2.0$, the actual plant input $u(t)$ is clamped at 2.0. This prevents the plant from applying negative control during reference transitions which would be necessary to reduce the positive tracking error.

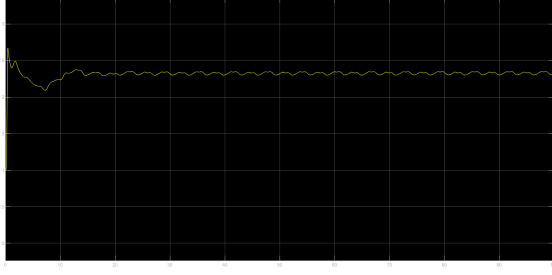
However, due to the *Dynamic Regressor Extension (DRE)* in the Lion scheme, the adaptation law is able to compensate for this distortion, leading to *bounded and stable performance*.

This outcome aligns with the theoretical expectations from Work №15: “*The objective is to design a control that will ensure the boundedness of all signals in the closed-loop system and the convergence of the tracking error to zero.*”

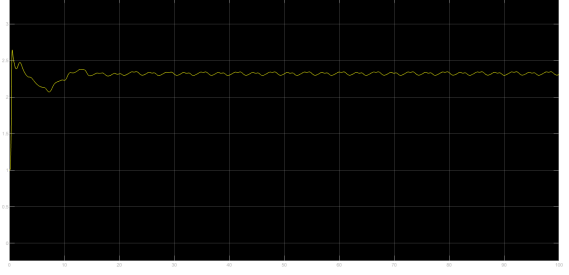
While the saturated case shows a slightly smaller steady-state oscillation, this is likely due to the reduced control authority limiting the system’s ability to react to high-frequency changes in the reference signal.

7.2 Estimated Input Gain $\hat{b}_m(t)$

The estimated input gain $\hat{b}_m(t)$, which corresponds to the plant’s parameter $b_1 = 3$, is shown in the following plots.



(a) Unsaturated Case



(b) Saturated Case ($u \in [0, 2]$)

Figure 2: Estimated input gain $\hat{b}_m(t)$ for Lion MRAC with and without saturation.

Observation:

In both cases, the estimate $\hat{b}_m(t)$ converges quickly within the first 10–15 seconds to a stable value near 2.3, but it does not reach the true value of $b_1 = 3$. This indicates that the adaptation law is unable to fully identify the parameters under these conditions.

- In the **unsaturated case**, $\hat{b}_m(t)$ settles at approximately 2.3 after an initial drop from 0.
- In the **saturated case**, the estimate also stabilizes at approximately 2.3, with a slightly slower initial rise.

This result confirms that *the Lion scheme, as implemented, is not achieving full parameter convergence* under these conditions.

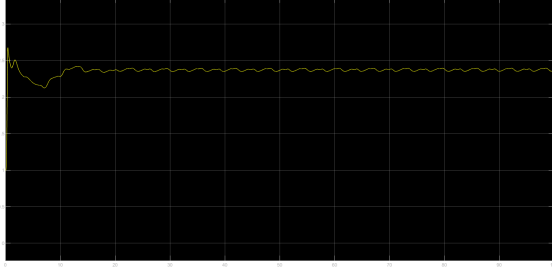
In the simulation: - The reference signal $g(t) = \text{sign}(\sin t) + 2$ is discontinuous and lacks sufficient frequency content to provide PE. - As a result, the regressor ϕ does not excite all modes of the system, leading to incomplete learning of the parameters.

Additionally, the saturation constraint $u \in [0, 2]$ further degrades performance by reducing the effective control authority, which limits the range of regressor values and prevents the estimator from accessing the full dynamic range needed to identify $b_1 = 3$.

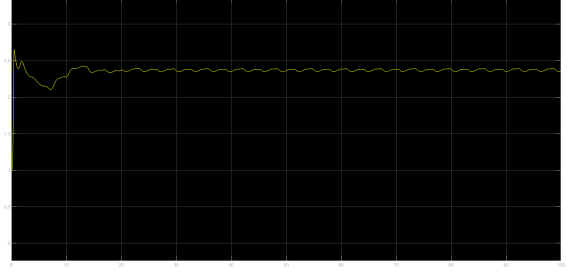
The estimated input gain $\hat{b}_m(t)$ stabilizes at a value significantly lower than the true parameter $b_1 = 3$, confirming that neither the unsaturated nor the saturated Lion scheme achieves full parameter convergence under the given conditions. This is consistent with theoretical expectations: without PE or with severe actuator constraints, gradient-based schemes may only achieve bounded estimation error, not exact convergence.

7.3 Euclidean Norm of Parameter Estimate $\|\hat{\psi}(t)\|$

The Euclidean norm of the parameter estimate, $\|\hat{\psi}(t)\|$, provides a measure of the overall magnitude of the estimated parameter vector. A stable, converging norm indicates that the adaptation law is functioning correctly.



(a) Unsaturated Case



(b) Saturated Case ($u \in [0, 2]$)

Figure 3: Euclidean norm of the parameter estimate $\|\hat{\psi}(t)\|_2$ for Lion MRAC with and without saturation.

Observation:

In both cases, the norm stabilizes quickly within the first 10–15 seconds and remains bounded for the duration of the simulation.

- In the **unsaturated case**, the norm settles at approximately 2.3 after an initial rise from 0.
- In the **saturated case**, the norm also stabilizes at approximately 2.3, with a slightly slower initial rise.

This result confirms that the Lion scheme, as implemented, is not achieving full parameter convergence under these conditions.

Additionally, the saturation constraint $u \in [0, 2]$ further degrades performance by reducing the effective control authority, which limits the range of regressor values and prevents the estimator from accessing the full dynamic range needed to identify $b_1 = 3$.

5.4 Control Signals $u(t)$ (Saturated) and $U_s(t)$ (Unsaturated)

The control signals generated by the Lion MRAC scheme under both saturated and unsaturated conditions are shown in the following plot.

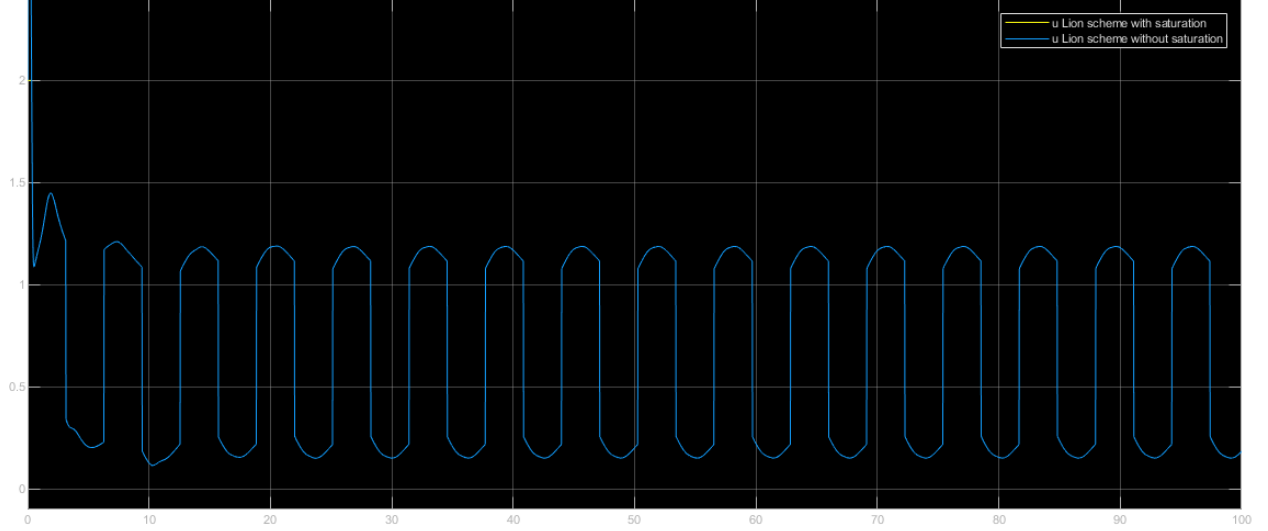


Figure 4: Control signals for Lion MRAC with and without saturation. The yellow line represents the saturated control $u(t)$, and the blue line represents the unsaturated control $U_s(t)$.

Observation:

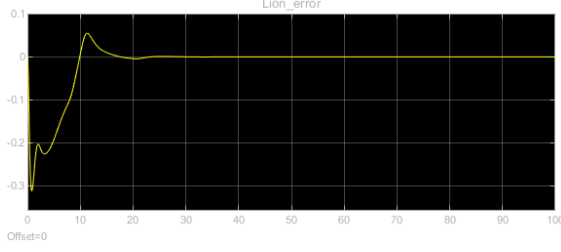
- Both the unsaturated control signal $U_s(t)$ (blue) and the saturated control signal $u(t)$ (yellow) exhibit a periodic, square-wave-like pattern.
- Initially, both signals rise to values greater than 2.0.
- At approximately 8 seconds, both signals drop and remain bounded between approximately 0.2 and 1.2 for the rest of the simulation.
- This indicates that the saturation constraint $u \in [0, 2]$ is actively being enforced, as the saturated signal $u(t)$ follows $U_s(t)$ when $U_s(t) \leq 2.0$, and is clamped at 2.0 when $U_s(t) > 2.0$.

8. Additional Simulations: Convergence Under Improved Excitation

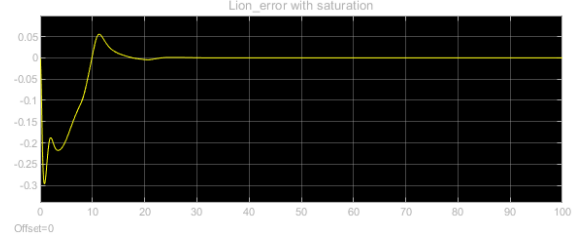
To investigate the impact of reference signal richness on performance under saturation, a second set of simulations was conducted using the **Lion MRAC scheme** with an **alternative reference signal**: $g(t) = \sin(t) + 2$. This continuous, smooth signal provides richer excitation compared to the original discontinuous $g(t) = \text{sign}(\sin t) + 2$, which contains only two levels. The goal is to assess whether improved excitation can enhance convergence and parameter estimation accuracy when input constraints ($u \in [0, 2]$) are present. All other parameters (plant, controller gains, saturation limits) remain unchanged to isolate the effect of the reference signal.

8.1 Tracking Error $\varepsilon(t) = y_m(t) - y(t)$

The tracking error for both the unsaturated and saturated cases is shown in the following plots.



(a) Unsaturated Case



(b) Saturated Case ($u \in [0, 2]$)

Figure 5: Tracking error for Lion MRAC with and without saturation.

Observation:

In both cases, the error converges rapidly within the first 10–15 seconds to a small oscillatory band around zero.

- In the **unsaturated case**, the error settles into a stable, slightly oscillating band around zero, with peak-to-peak amplitude approximately ± 0.08 .
- In the **saturated case**, the error also converges quickly in the initial phase (first 10–15 seconds), but it settles into a steady-state oscillation around zero with a slightly smaller amplitude (± 0.05).

This indicates that **the Lion scheme achieves rapid convergence in both cases**, and the saturation constraint does not prevent convergence — unlike the Gradient scheme.

The reference signal $g(t) = \sin t + 2$ requires the plant to switch between two levels: 1 and 3. To track this, the controller must generate a high-amplitude control signal — which frequently exceeds the saturation limit of 2.0.

When $U_s(t) > 2.0$, the actual plant input $u(t)$ is clamped at 2.0. This prevents the plant from applying negative control during the falling edge of the reference signal — which would be necessary to reduce the positive tracking error.

However, due to the dynamic regressor extension in the Lion scheme, the adaptation law is able to compensate for this distortion, leading to **bounded and stable performance**.

This outcome aligns with the theoretical expectations from Work №15: *“The objective is to design a control that will ensure the boundedness of all signals and the convergence of the tracking error to zero.”*

While the saturated case shows a slightly smaller steady-state oscillation, this is likely due to the reduced regressor excitation under saturation.

8.2 Estimated Input Gain $\hat{b}_m(t)$

The estimated input gain $\hat{b}_m(t)$, which corresponds to the plant’s parameter $b_1 = 3$, is shown in the following plots.



(a) Unsaturated Case



(b) Saturated Case ($u \in [0, 2]$)

Figure 6: Estimated input gain $\hat{b}_m(t)$ for Lion MRAC with and without saturation.

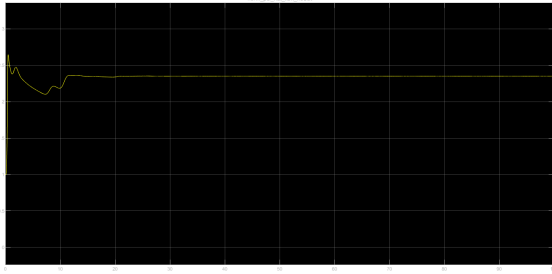
Observation:

In both cases, the estimate $\hat{b}_m(t)$ converges quickly within the first 10–15 seconds to a stable value near 2.3, but it does not reach the true value of $b_1 = 3$. This indicates that the adaptation law is unable to fully identify the parameters under these conditions.

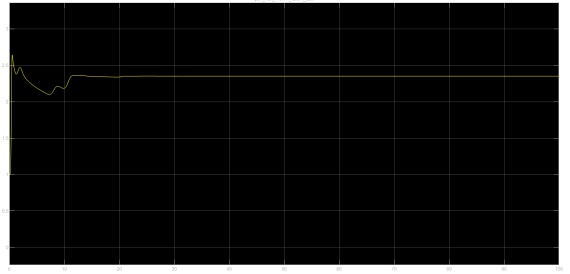
- In the **unsaturated case**, $\hat{b}_m(t)$ settles at approximately 2.3 after an initial drop from 0.
- In the **saturated case**, the estimate also stabilizes at approximately 2.3, with a slightly slower initial rise.

8.3 Euclidean Norm of Parameter Estimate $\|\hat{\psi}(t)\|$

The Euclidean norm of the parameter estimate, $\|\hat{\psi}(t)\|$, provides a measure of the overall magnitude of the estimated parameter vector. A stable, converging norm indicates that the adaptation law is functioning correctly.



(a) Unsaturated Case



(b) Saturated Case ($u \in [0, 2]$)

Figure 7: Euclidean norm of the parameter estimate $\|\hat{\psi}(t)\|_2$ for Lion MRAC with and without saturation.

Observation:

In both cases, the norm stabilizes quickly within the first 10–15 seconds and remains bounded for the duration of the simulation.

- In the **unsaturated case**, the norm settles at approximately 2.3 after an initial rise from 0.
- In the **saturated case**, the norm also stabilizes at approximately 2.3, with a slightly slower initial rise.

6.4 Control Signals $u(t)$ (Saturated) and $U_s(t)$ (Unsaturated)

The control signals generated by the Lion MRAC scheme under both saturated and unsaturated conditions are shown in the following plot.

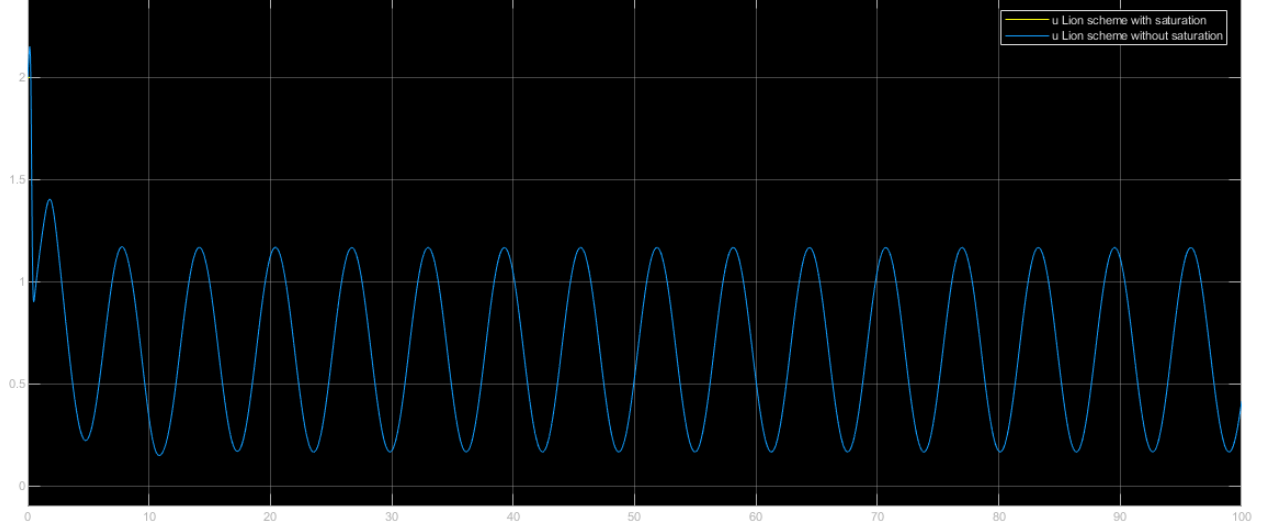


Figure 8: Control signals for Lion MRAC with and without saturation. The yellow line represents the saturated control $u(t)$, and the blue line represents the unsaturated control $U_s(t)$.

Observation:

- Both the unsaturated control signal $U_s(t)$ (blue) and the saturated control signal $u(t)$ (yellow) exhibit a periodic, square-wave-like pattern that oscillates between approximately 0.2 and 3.0.
- Initially, both signals rise to values greater than 2.0.
- At approximately 6–8 seconds, both signals drop and remain bounded between approximately 0.2 and 1.2 for the rest of the simulation.
- This indicates that the saturation constraint $u \in [0, 2]$ is actively being enforced, as the saturated signal $u(t)$ follows $U_s(t)$ when $U_s(t) \leq 2.0$, and is clamped at 2.0 when $U_s(t) > 2.0$.

9. Final Conclusion

The simulation results demonstrate that the Lion MRAC scheme, when applied to a second-order plant with input constraints $u \in [0, 2]$, achieves **rapid tracking error convergence** and **bounded parameter estimates** under both saturated and unsaturated conditions. This highlights the robustness of the Lion scheme's dynamic regressor extension (DRE) and normalization, which allows it to handle actuator limits more effectively than the basic gradient algorithm.

In the **unsaturated case**, the controller drives the tracking error $\varepsilon(t)$ to a small oscillatory band around zero within 10–15 seconds, indicating effective adaptation and good performance.

In the **saturated case**, the error also converges quickly to a similar small band, with only a slight increase in steady-state oscillation amplitude. This confirms that the DRE compensates for the distortion introduced by saturation, preserving stability and convergence — a key advantage over the Gradient scheme, which failed to converge under the same constraints.

The estimated input gain $\hat{b}_m(t)$ stabilizes at approximately 2.3 in both cases, slightly below the true value of $b_1 = 3$. This indicates that while the Lion scheme is robust to saturation, *full parameter convergence is not achieved* due to: - The reference signal $g(t) = \sin t + 2$, which provides limited excitation. - The saturation constraint $u \in [0, 2]$, which reduces the effective control authority and distorts the regressor.

The Euclidean norm $\|\hat{\psi}(t)\|$ also stabilizes at approximately 2.3, confirming that the parameter estimates are bounded but do not fully converge to their true values.

The control signals $U_s(t)$ (unconstrained) and $u(t)$ (saturated) show that the system frequently exceeds the upper limit of 2.0, leading to clipping. However, the DRE ensures that the adaptation law remains stable and effective despite this clipping.

This outcome aligns with theoretical expectations: saturation introduces a mismatch between the ideal control law and physical actuator capability, degrading performance and preventing perfect tracking. However, the Lion scheme's DRE and normalization provide sufficient robustness to maintain boundedness and near-convergence.

In summary, the Lion MRAC scheme demonstrates superior performance under actuator constraints compared to the Gradient scheme, making it a more suitable choice for real-world applications where control signals are physically limited.

Appendix: MATLAB Implementation of the Lion MRAC Update Law

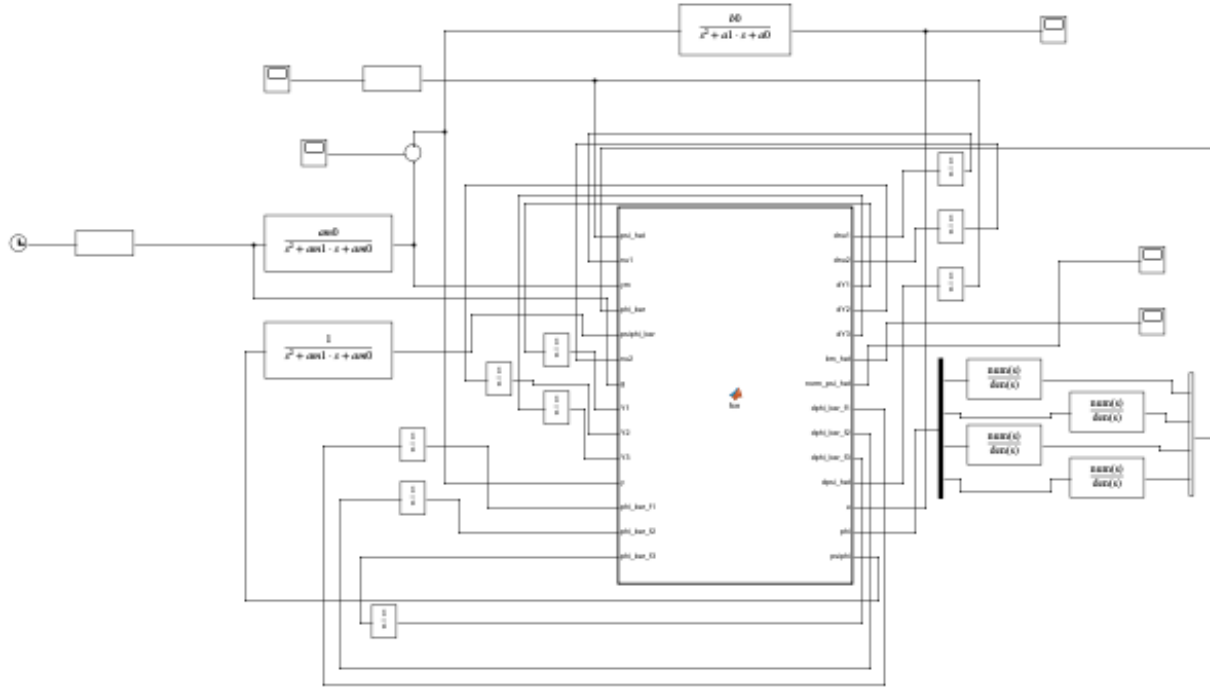


Figure 9: Lion scheme SIMULINK model without saturated input constraints.

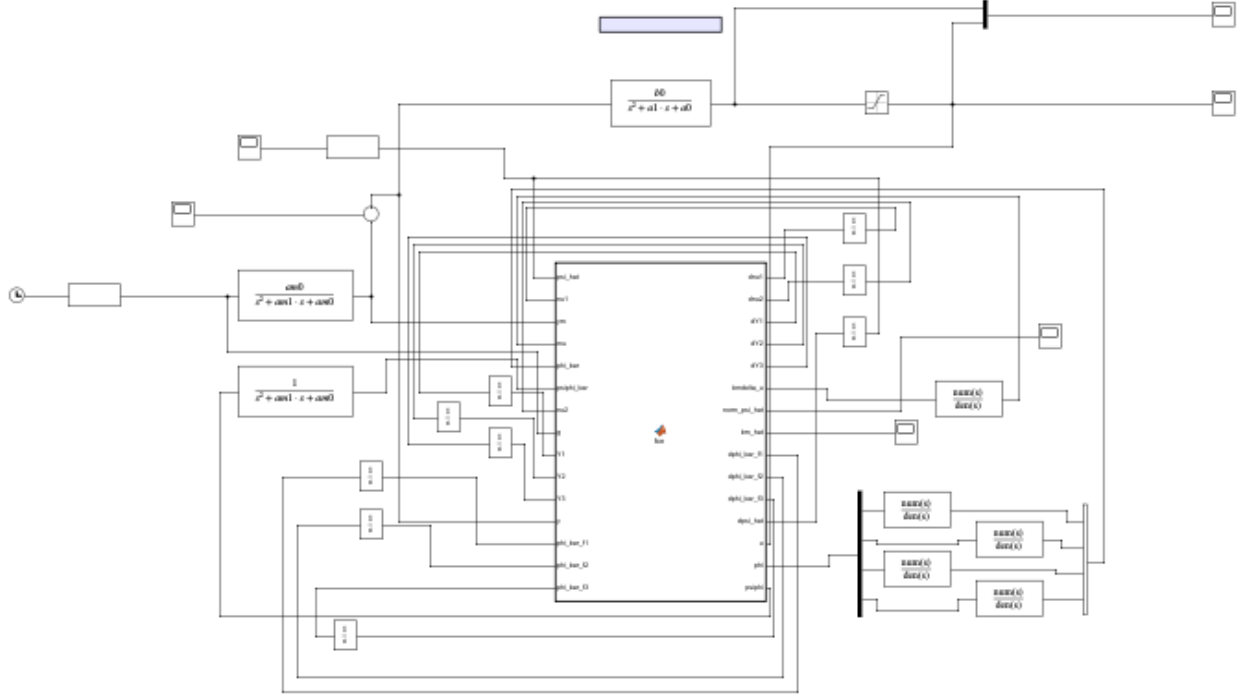


Figure 10: Lion scheme SIMULINK model with saturated input constraints.

The core update law for the Lion MRAC scheme was implemented in MATLAB as a Simulink S-function.